

ASSOCIATION RULES

With JMP software

مقدمه

اطلاعات مربوط به یک کسب‌وکار واقعی مرتبط با فروش کتاب در اختیارمان قرار داده شده است. در این تمرین قرار است یکسری تحلیل روی کتاب‌ها، ناشران، نویسندگان و مشتریان داشته باشیم و روابط ارتباطی بین فروش کتاب‌ها، نویسنده‌ها و ... را نیز مورد بررسی قرار دهیم.

ناهید قلی‌زاده

دوره جامع تربیت تحلیل‌گر داده

- **۲۰ نویسنده با بیشترین تعداد کتاب چاپ شده**

	Book-Author	Num Of Published Book
1	Agatha Christie	631
2	William Shakespe...	567
3	Stephen King	524
4	Ann M. Martin	423
5	Carolyn Keene	373
6	Francine Pascal	372
7	Isaac Asimov	330
8	Nora Roberts	315
9	Barbara Cartland	307
10	Charles Dickens	302
11	Not Applicable (...	285
12	R. L. Stine	282
13	Mark Twain	231
14	Jane Austen	223
15	Terry Pratchett	220
16	Mary Higgins Clark	218
17	Piers Anthony	217
18	Marion Zimmer B...	216
19	Janet Dailey	214
20	Franklin W. Dixon	204

- **۲۰ نویسنده با بیشترین تعداد کتاب فروخته شده**

	Book-Author	Num Of Sold Book
1	Stephen King	10053
2	Nora Roberts	8429
3	John Grisham	6010
4	James Patterson	5845
5	Mary Higgins Clark	4777
6	Dean R. Koontz	4313
7	Tom Clancy	4036
8	Danielle Steel	3726
9	Sue Grafton	3456
10	Janet Evanovich	3350
11	Anne Rice	3030
12	Michael Crichton	2921
13	J. K. Rowling	2908
14	JOHN GRISHAM	2808
15	V.C. Andrews	2785
16	Sandra Brown	2663
17	Nicholas Sparks	2650
18	R. L. Stine	2606
19	Rich Shapero	2502
20	Patricia Daniels C...	2461

- **۲۰ ناشر با بیشترین تعداد کتاب فروخته شده**

	Publisher	Num Of Sold Book
1	Ballantine Books	34722
2	Pocket	31988
3	Berkley Publishin...	28614
4	Warner Books	25506
5	Harlequin	25020
6	Bantam Books	23569
7	Bantam	20007
8	Signet Book	19155
9	Avon	17352
10	Penguin Books	17032
11	Jove Books	15177
12	Silhouette	14182
13	Dell	13916
14	Scholastic	13661
15	Perennial	13466
16	Fawcett Books	12901
17	Dell Publishing C...	12730
18	HarperTorch	12081
19	Vintage Books USA	11426
20	Simon &amp, Sc...	10315

- **۲۰ کتاب با بیشترین تعداد فروش**

	Book-Title	Num Of Sales
1	Wild Animus	2502
2	The Lovely Bones...	1295
3	The Da Vinci Code	898
4	A Painted House	838
5	The Nanny Diarie...	828
6	Bridget Jones's D...	815
7	The Secret Life of...	774
8	Divine Secrets of ...	740
9	The Red Tent (Be...	723
10	Angels &amp, De...	670
11	Life of Pi	664
12	Snow Falling on ...	662
13	The Summons	655
14	The Notebook	650
15	The Testament	617
16	House of Sand a...	588
17	Where the Heart ...	585
18	The Pelican Brief	581
19	Harry Potter and ...	575
20	Summer Sisters	573

- **۲۰ کشور با بیشترین تعداد سفارش کتاب**

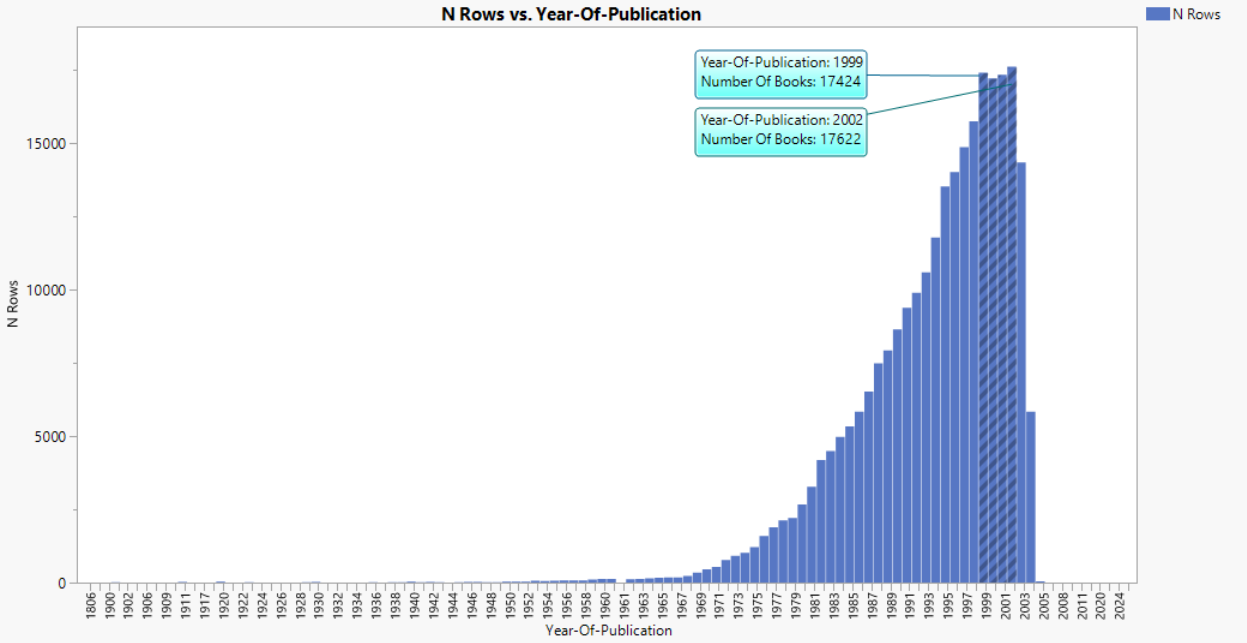
	Country	Number Of Book Orders
1	usa	11197
2	germany	1783
3	united kingdom	1778
4	canada	1342
5	spain	1131
6	australia	1110
7	italy	931
8	france	583
9	switzerland	407
10	netherlands	405
11	portugal	331
12	new zealand	290
13	sweden	182
14	belgium	142
15	austria	135
16	denmark	120
17	brazil	109
18	ireland	109
19	malaysia	109
20	finland	103

- **تحلیل سال انتشار کتاب‌ها**

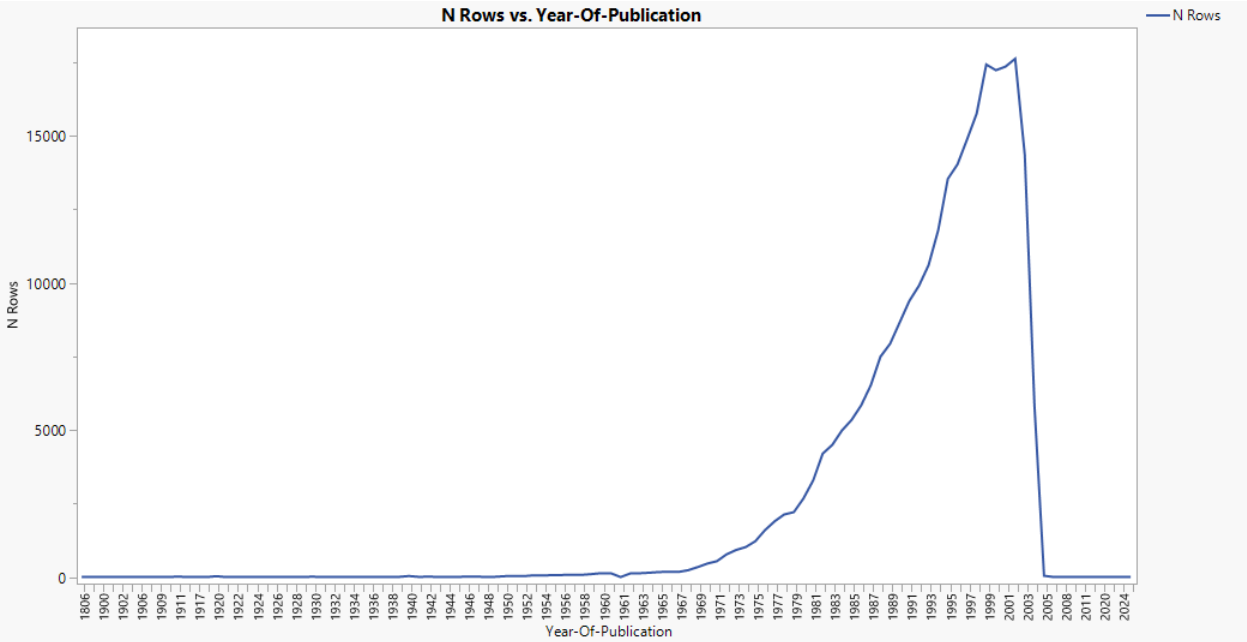
	Year-Of-Publication	Number Of Books
1	2002	17622
2	1999	17424
3	2001	17354
4	2000	17230
5	1998	15760
6	1997	14880
7	2003	14354
8	1996	14026
9	1995	13539
10	1994	11789
11	1993	10602
12	1992	9905
13	1991	9387
14	1990	8657
15	1989	7931
16	1988	7491
17	1987	6525
18	1986	5839
19	2004	5838
20	1985	5342

کتاب‌ها در ۱۱۰ سال متفاوت از سال ۱۸۰۶ تا سال ۲۰۲۶ به چاپ رسیده و منتشر شده‌اند. در بین این ۱۱۰ سال مختلف، همان طور که در جدول بالا مشاهده می‌کنید، سال ۲۰۰۲ با ۱۷۶۲۲ مورد کتاب در رتبه نخست و سال ۱۹۸۵ با ۵۳۴۲ کتاب در رتبه ۲۰ جدول قرار گرفته‌اند.

در نمودار زیر مشهود است که بیشتر کتاب‌ها متعلق به سال‌های ۱۹۹۹ تا ۲۰۰۲ هستند.



از طرف دیگر، تعداد کتاب‌های متعلق به سال‌های ۲۰۰۳ تا ۲۰۰۵ دچار افت شدیدی شده‌اند.

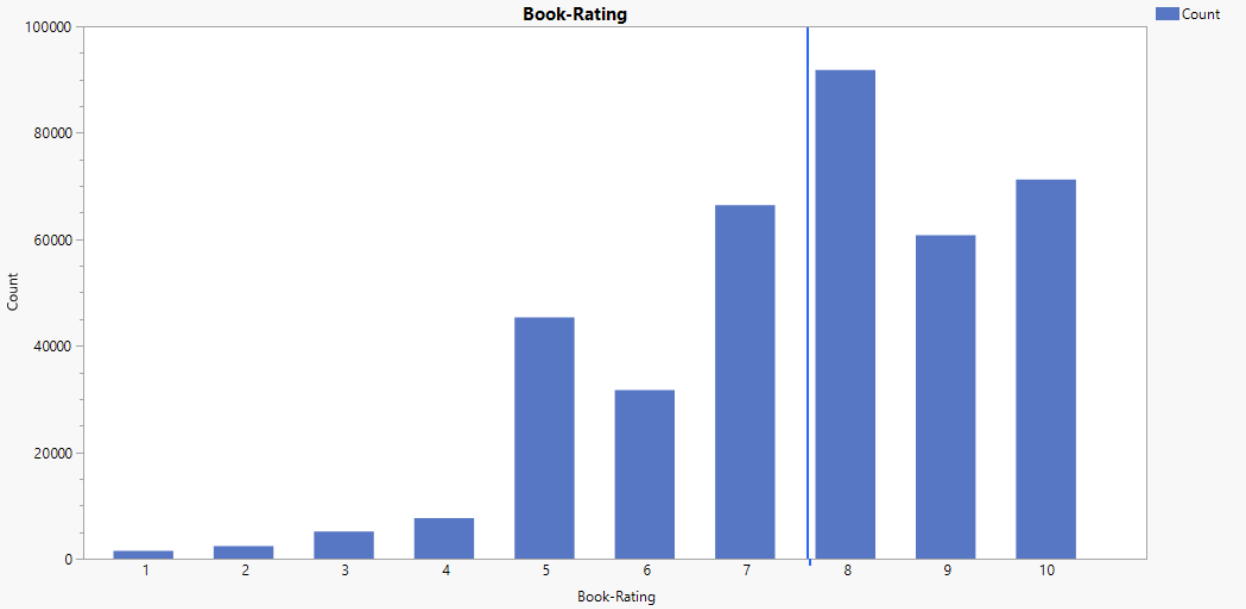


➢ تحلیل نمره‌های داده شده به کتاب‌ها

خریداران به کتاب‌ها از ۱ تا ۱۰ امتیازهای مختلفی داده‌اند. بیشترین امتیاز متعلق به نمره ۸ است که ۱۰۳۷۳۷ نفر به کتاب‌ها این امتیاز را داده‌اند.

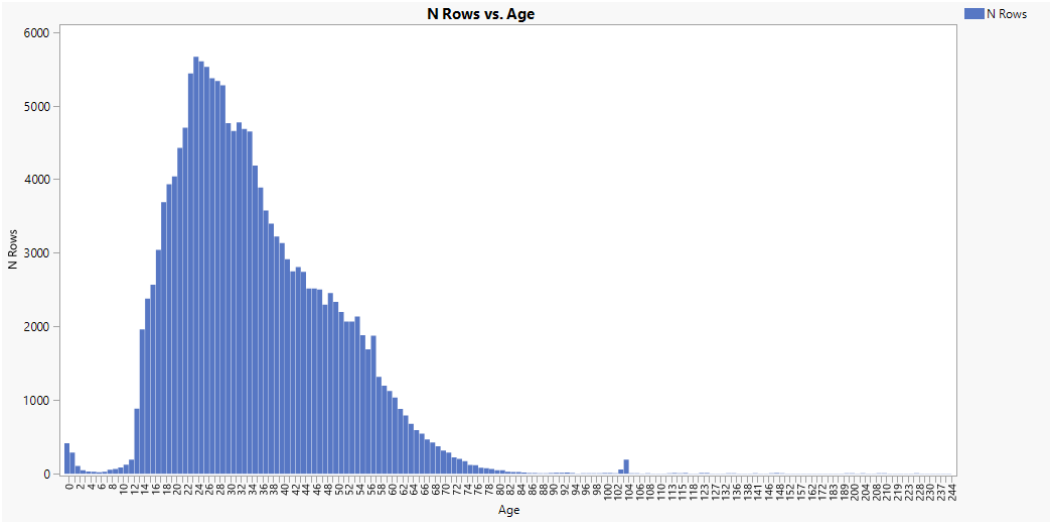
میانگین امتیازها نیز ۷.۶۳ می‌باشد.

	N Rows	Mean(Book-Rating)
1	383850	7.63

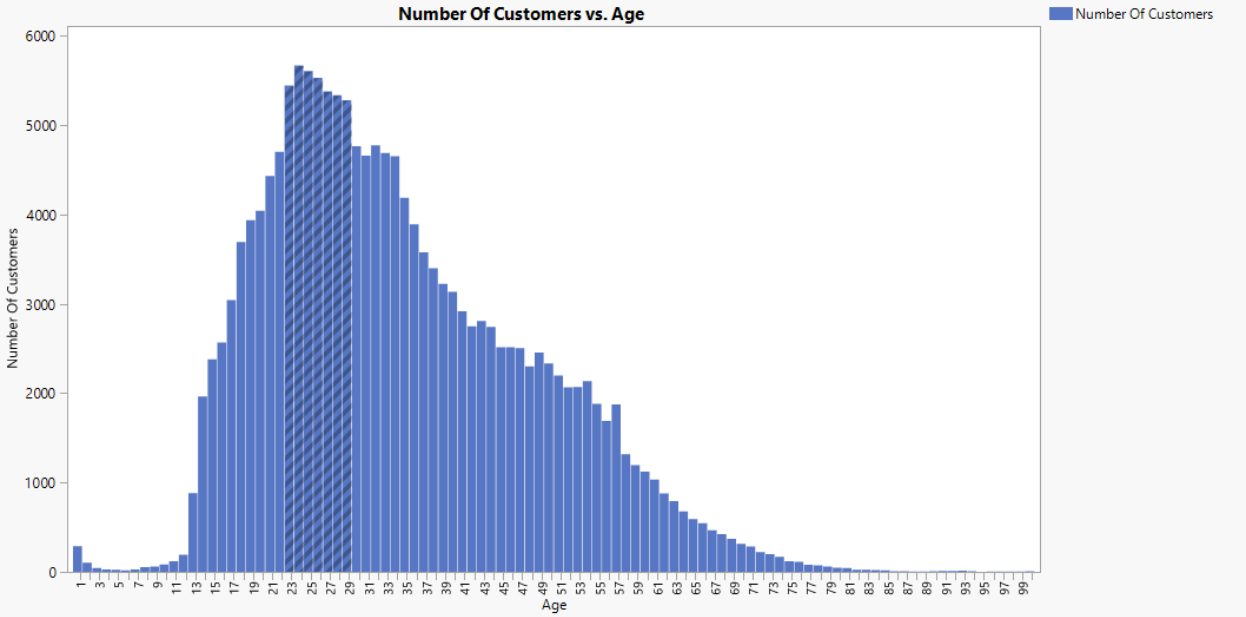


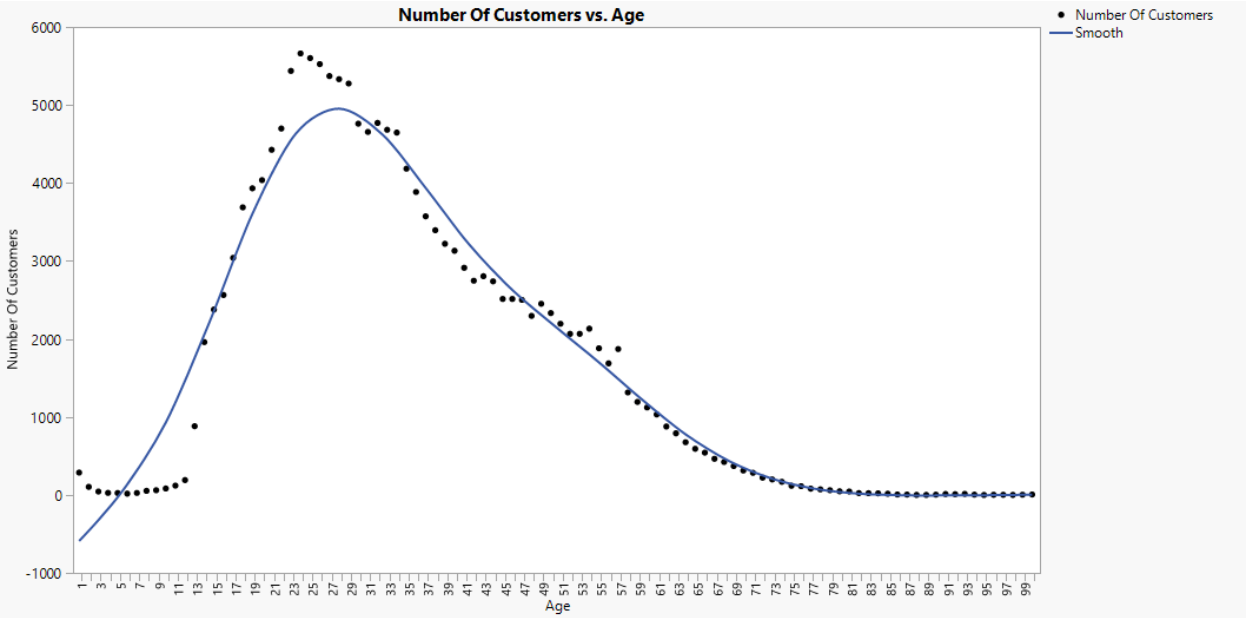
➢ تحلیل سن مشتریان

ابتدا داده‌های مربوط به سن مشتریان را پاکسازی می‌کنیم و آن‌هایی را که مقادیر غیرعددی یا Null خورده بودند را حذف می‌کنیم. در این صورت نموداری مشابه نمودار زیر خواهیم داشت.



اما اگر بازه سنی مجاز را ببین ۰ تا ۱۰۰ سال در نظر بگیریم، نمودار به این شکل خواهد بود.





همان‌طور که در دو نمودار بالا مشهود است، اکثر رنج سنی افرادی که کتاب خریداری کرده‌اند متعلق به گروه سنی بین ۲۳ تا ۲۹ سال است که بیشترین آن‌ها یعنی ۵۶۶۴ نفر ۲۱ سال سن داشته‌اند.

- **۲۰ قانون با بالاترین Lift برای کتاب‌ها**

اگر چه فضای قوانین ۱←۱ متناهی است، اما استخراج عملی آن مستلزم تعیین آستانه‌ی حداقل Support است. قوانین با بیشترین Lift در چارچوب زیر انتخاب شده‌اند:

Minimum Support = 0.001

Minimum Confidence = 0.04

- با احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift
1	Key of Knowledge (Key Trilogy (Paperback))	Key of Light (Key Trilogy (Paperback))	81%	549.9
2	Key of Light (Key Trilogy (Paperback))	Key of Knowledge (Key Trilogy (Paperback))	70%	549.9
3	Nicolae: The Rise of Antichrist (Left Behind No. 3)	Tribulation Force: The Continuing Drama of Those Left Behin...	76%	385.6
4	Tribulation Force: The Continuing Drama of Those Left Behind (Left...	Nicolae: The Rise of Antichrist (Left Behind No. 3)	53%	385.6
5	The Amber Spyglass (His Dark Materials, Book 3)	The Subtle Knife (His Dark Materials, Book 2)	77%	318.6
6	The Subtle Knife (His Dark Materials, Book 2)	The Amber Spyglass (His Dark Materials, Book 3)	48%	318.6
7	Hot Six : A Stephanie Plum Novel (A Stephanie Plum Novel)	High Five (A Stephanie Plum Novel)	59%	242.6
8	High Five (A Stephanie Plum Novel)	Hot Six : A Stephanie Plum Novel (A Stephanie Plum Novel)	53%	242.6
9	High Five (A Stephanie Plum Novel)	Four To Score (A Stephanie Plum Novel)	59%	241.2
10	Four To Score (A Stephanie Plum Novel)	High Five (A Stephanie Plum Novel)	59%	241.2
11	The Return of the King (The Lord of the Rings, Part 3)	The Two Towers (The Lord of the Rings, Part 2)	66%	240
12	The Two Towers (The Lord of the Rings, Part 2)	The Return of the King (The Lord of the Rings, Part 3)	54%	240
13	Four To Score (A Stephanie Plum Novel)	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie P...	56%	228.4
14	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie Plum ...	Four To Score (A Stephanie Plum Novel)	56%	228.4
15	C Is for Corpse (Kinsey Millhone Mysteries (Paperback))	B Is for Burglar (Kinsey Millhone Mysteries (Paperback))	60%	224.2
16	B Is for Burglar (Kinsey Millhone Mysteries (Paperback))	C Is for Corpse (Kinsey Millhone Mysteries (Paperback))	43%	224.2
17	Hot Six : A Stephanie Plum Novel (A Stephanie Plum Novel)	Four To Score (A Stephanie Plum Novel)	55%	223.1
18	Four To Score (A Stephanie Plum Novel)	Hot Six : A Stephanie Plum Novel (A Stephanie Plum Novel)	48%	223.1
19	High Five (A Stephanie Plum Novel)	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie P...	53%	218.4
20	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie Plum ...	High Five (A Stephanie Plum Novel)	53%	218.4

- بدون احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift
1	The Two Towers (The Lord of the Rings, Part 2)	The Fellowship of the Ring (The Lord of the Rings, Part 1)	56%	186.3437329
2	The Fellowship of the Ring (The Lord of the Rings, Part 1)	The Two Towers (The Lord of the Rings, Part 2)	37%	186.3437329
3	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Prisoner of Azkaban (Book 3)	58%	143.76244194
4	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Goblet of Fire (Book 4)	51%	143.76244194
5	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Chamber of Secrets (Book 2)	56%	119.21981945
6	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Prisoner of Azkaban (Book 3)	48%	119.21981945
7	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Chamber of Secrets (Book 2)	55%	117.45397904
8	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Goblet of Fire (Book 4)	42%	117.45397904
9	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	114.91498249
10	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Prisoner of Azkaban (Book 3)	46%	114.91498249
11	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Goblet of Fire (Book 4)	40%	113.84655807
12	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	113.84655807
13	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Goblet of Fire (Book 4)	40%	113.34554072
14	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Order of the Phoenix (Book 5)	35%	113.34554072
15	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Chamber of Secrets (Book 2)	52%	110.35060234
16	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Sorcerer's Stone (Book 1)	28%	110.35060234
17	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Prisoner of Azkaban (Book 3)	38%	94.219393227
18	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Order of the Phoenix (Book 5)	29%	94.219393227
19	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Chamber of Secrets (Book 2)	36%	75.397528458
20	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Order of the Phoenix (Book 5)	23%	75.397528458

➢ ۲۰ قانون با بالاترین Confidence برای کتاب‌ها

برای مقایسه منصفانه با شرایط قسمت قبل، مقدار Support یکسان در نظر گرفته شده است.

Minimum Support = 0.001

Minimum Confidence = 0.04

- با احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift
1	Key of Knowledge (Key Trilogy (Paperback))	Key of Light (Key Trilogy (Paperback))	81%	549.9
2	The Amber Spyglass (His Dark Materials, Book 3)	The Subtle Knife (His Dark Materials, Book 2)	77%	318.6
3	Nicolae: The Rise of Antichrist (Left Behind No. 3)	Tribulation Force: The Continuing Drama of Those Left Behind (Left...	76%	385.6
4	Key of Light (Key Trilogy (Paperback))	Key of Knowledge (Key Trilogy (Paperback))	70%	549.9
5	The Return of the King (The Lord of the Rings, Part 3)	The Two Towers (The Lord of the Rings, Part 2)	66%	240
6	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Chamber of Secrets (Book 2)	63%	106.9
7	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Chamber of Secrets (Book 2)	62%	104.2
8	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Prisoner of Azkaban (Book 3)	61%	134
9	The Two Towers (The Lord of the Rings, Part 2)	The Fellowship of the Ring (The Lord of the Rings, Part 1)	61%	157.9
10	Shopaholic Takes Manhattan (Summer Display Opportunity)	Confessions of a Shopaholic (Summer Display Opportunity)	60%	157.9
11	C Is for Corpse (Kinsey Millhone Mysteries (Paperback))	B Is for Burglar (Kinsey Millhone Mysteries (Paperback))	60%	224.2
12	Hot Six : A Stephanie Plum Novel (A Stephanie Plum Novel)	High Five (A Stephanie Plum Novel)	59%	242.6
13	High Five (A Stephanie Plum Novel)	Four To Score (A Stephanie Plum Novel)	59%	241.2
14	Tribulation Force: The Continuing Drama of Those Left Behind (Left...	Left Behind: A Novel of the Earth's Last Days (Left Behind No. 1)	59%	171.4
15	Four To Score (A Stephanie Plum Novel)	High Five (A Stephanie Plum Novel)	59%	241.2
16	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie Plum ...	Two for the Dough	58%	200.2
17	The Return of the King (The Lord of the Rings, Part 3)	The Fellowship of the Ring (The Lord of the Rings, Part 1)	57%	147.5
18	Heaven and Earth (Three Sisters Island Trilogy)	Dance upon the Air (Three Sisters Island Trilogy)	56%	200.6
19	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie Plum ...	Four To Score (A Stephanie Plum Novel)	56%	228.4
20	Four To Score (A Stephanie Plum Novel)	Three To Get Deadly : A Stephanie Plum Novel (A Stephanie Plum ...	56%	228.4

- بدون احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift
1	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Prisoner of Azkaban (Book 3)	58%	143.76244194
2	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Chamber of Secrets (Book 2)	56%	119.21981945
3	The Two Towers (The Lord of the Rings, Part 2)	The Fellowship of the Ring (The Lord of the Rings, Part 1)	56%	186.3437329
4	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Chamber of Secrets (Book 2)	55%	117.45397904
5	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Chamber of Secrets (Book 2)	52%	110.35060234
6	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Goblet of Fire (Book 4)	51%	143.76244194
7	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Prisoner of Azkaban (Book 3)	48%	119.21981945
8	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Prisoner of Azkaban (Book 3)	46%	114.91498249
9	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Goblet of Fire (Book 4)	42%	117.45397904
10	Harry Potter and the Sorcerer's Stone (Book 1)	Harry Potter and the Goblet of Fire (Book 4)	40%	113.84655807
11	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Goblet of Fire (Book 4)	40%	113.34554072
12	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Prisoner of Azkaban (Book 3)	38%	94.219393227
13	The Fellowship of the Ring (The Lord of the Rings, Part 1)	The Two Towers (The Lord of the Rings, Part 2)	37%	186.3437329
14	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Chamber of Secrets (Book 2)	36%	75.397528458
15	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Order of the Phoenix (Book 5)	35%	113.34554072
16	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	Harry Potter and the Chamber of Secrets (Book 2)	29%	61.952034812
17	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	114.91498249
18	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Order of the Phoenix (Book 5)	29%	94.219393227
19	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	113.84655807
20	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	29%	61.952034812

➢ ۲۰ قانون با بالاترین LHS Support برای کتاب‌ها

- با احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift	LHS Support
1	Wild Animus	The Lovely Bones: A Novel	10%	6.768	2.72%
2	Wild Animus	The Da Vinci Code	8%	8.17	2.72%
3	Wild Animus	Bridget Jones's Diary	7%	8.73	2.72%
4	Wild Animus	Divine Secrets of the Ya-Ya Sisterhood: A Novel	6%	7.945	2.72%
5	Wild Animus	The Secret Life of Bees	6%	7.684	2.72%
6	Wild Animus	The Nanny Diaries: A Novel	6%	6.818	2.72%
7	Wild Animus	Angels &amp, Demons	6%	8.413	2.72%
8	Wild Animus	A Painted House	6%	6.753	2.72%
9	Wild Animus	Life of Pi	6%	7.947	2.72%
10	Wild Animus	The Notebook	6%	8.389	2.72%
11	Wild Animus	Snow Falling on Cedars	6%	7.756	2.72%
12	Wild Animus	The Girls' Guide to Hunting and Fishing	5%	9.008	2.72%
13	Wild Animus	The Firm	5%	8.926	2.72%
14	Wild Animus	House of Sand and Fog	5%	8.401	2.72%
15	Wild Animus	The Red Tent (Bestselling Backlist)	5%	6.774	2.72%
16	Wild Animus	The Pelican Brief	5%	8.434	2.72%
17	Wild Animus	Summer Sisters	5%	8.322	2.72%
18	Wild Animus	A Time to Kill	5%	8.161	2.72%
19	Wild Animus	Harry Potter and the Chamber of Secrets (Book 2)	5%	8.109	2.72%
20	Wild Animus	The Joy Luck Club	5%	7.977	2.72%

- بدون احتساب امتیاز صفر

	Condition	Consequent	Confidence	Lift	LHS Support
1	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Prisoner of Azkaban (Book 3)	48%	119.21981945	0.47%
2	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Goblet of Fire (Book 4)	42%	117.45397904	0.47%
3	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	29%	61.952034812	0.47%
4	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Sorcerer's Stone (Book 1)	28%	110.35060234	0.47%
5	Harry Potter and the Chamber of Secrets (Book 2)	Harry Potter and the Order of the Phoenix (Book 5)	23%	75.397528458	0.47%
6	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	Harry Potter and the Chamber of Secrets (Book 2)	29%	61.952034812	0.46%
7	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	Harry Potter and the Prisoner of Azkaban (Book 3)	23%	57.589734677	0.46%
8	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Chamber of Secrets (Book 2)	56%	119.21981945	0.40%
9	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Goblet of Fire (Book 4)	51%	143.76244194	0.40%
10	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	114.91498249	0.40%
11	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Order of the Phoenix (Book 5)	29%	94.219393227	0.40%
12	Harry Potter and the Prisoner of Azkaban (Book 3)	Harry Potter and the Sorcerer's Stone (Harry Potter (Paperback))	27%	57.589734677	0.40%
13	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Prisoner of Azkaban (Book 3)	58%	143.76244194	0.36%
14	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Chamber of Secrets (Book 2)	55%	117.45397904	0.36%
15	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Order of the Phoenix (Book 5)	35%	113.34554072	0.36%
16	Harry Potter and the Goblet of Fire (Book 4)	Harry Potter and the Sorcerer's Stone (Book 1)	29%	113.84655807	0.36%
17	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Goblet of Fire (Book 4)	40%	113.34554072	0.31%
18	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Prisoner of Azkaban (Book 3)	38%	94.219393227	0.31%
19	Harry Potter and the Order of the Phoenix (Book 5)	Harry Potter and the Chamber of Secrets (Book 2)	36%	75.397528458	0.31%
20	The Fellowship of the Ring (The Lord of the Rings, Part 1)	The Two Towers (The Lord of the Rings, Part 2)	37%	186.3437329	0.30%

- **مقایسه میانگین نمرات ۱۰ کتاب با بالاترین Support با میانگین همه کتاب‌ها** (بدون احتساب امتیاز صفر)
- لیست ۱۰ کتاب با بالاترین support

	Item Set	Support
1	The Lovely Bones: A Novel	1.04%
2	Wild Animus	0.85%
3	The Da Vinci Code	0.73%
4	The Secret Life of Bees	0.59%
5	The Nanny Diaries: A Novel	0.57%
6	The Red Tent (Bestselling Backlist)	0.56%
7	Bridget Jones's Diary	0.54%
8	A Painted House	0.53%
9	Life of Pi	0.49%
10	Divine Secrets of the Ya-Ya Sisterhood: A Novel	0.47%

- لیست نمرات ۱۰ کتاب با بالاترین support

	Item Set	Support	N Rows	Sum(Book-Rating)
1	The Lovely Bones: A Novel	1.04%	707	5787
2	Wild Animus	0.85%	581	2551
3	The Da Vinci Code	0.73%	494	4169
4	The Secret Life of Bees	0.59%	406	3442
5	The Nanny Diaries: A Novel	0.57%	393	2923
6	The Red Tent (Bestselling Backlist)	0.56%	383	3134
7	Bridget Jones's Diary	0.54%	377	2875
8	A Painted House	0.53%	366	2708
9	Life of Pi	0.49%	336	2715
10	Divine Secrets of the Ya-Ya Sisterhood: A Novel	0.47%	323	2544

- میانگین نمرات ۱۰ کتاب با بالاترین support

Summary Statistics

Mean	3284.8
Std Dev	1007.4015
Std Err Mean	318.56832
Upper 95% Mean	4005.4516
Lower 95% Mean	2564.1484
N	10
N Missing	0

- میانگین نمرات ۱۰ کتاب با بالاترین support

Summary Statistics

Mean	21.594295
Std Dev	71.475601
Std Err Mean	0.1941233
Upper 95% Mean	21.974773
Lower 95% Mean	21.213817
N	135569
N Missing	0

- نتیجه‌گیری

مقدار میانگین امتیاز کل کتاب‌هایی که به آن‌ها امتیاز تعلق گرفته است، ۲۱ بوده در حالی که ۱۰ کتاب برتر با بالاترین Support، میانگین امتیازشان ۳۲۸۴ شده است. مشخصا کتاب‌هایی که کاربران امتیاز بالایی به آن‌ها می‌دهند، بیشتر مورد علاقه‌شان بوده و احتمال خریدشان نیز بالاتر است.

- **۲۰ قانون با بالاترین Lift برای نویسندگان**

- Minimum Support = 0.001
- Minimum Confidence = 0.5

	Condition	Consequent	Confidence	Lift
1	Tim F. Lahaye	Tim LaHaye	73%	269.3
2	Tim F. Lahaye	Tim Lahaye	78%	127.9
3	Tim LaHaye	Tim Lahaye	74%	121
4	Sophie Kinsella	SOPHIE KINSELLA	51%	107.5
5	DIANE MOTT DAVIDS...	Diane Mott David...	57%	106.9
6	Jennifer Blake	Johanna Lindsey	53%	89.58
7	Janelle Taylor	Johanna Lindsey	52%	87.22
8	Janelle Taylor	Debbie Macomber	53%	87.12
9	Jennifer Blake	Debbie Macomber	51%	84.04
10	Jane Feather	Johanna Lindsey	50%	84.04
11	R.L. Stine	R. L. Stine	52%	74.41
12	Janelle Taylor	Fern Michaels	55%	71.53
13	V. C. Andrews	V.C. Andrews	61%	70.11
14	Jennifer Blake	Jude Deveraux	59%	62.68
15	Kat Martin	Jude Deveraux	58%	60.91
16	Janelle Taylor	Jude Deveraux	56%	58.81
17	Jane Feather	Jude Deveraux	55%	58.43
18	Joan Johnston	Jude Deveraux	55%	57.91
19	Kat Martin	Catherine Coulter	63%	57.35
20	SANDRA BROWN	Sandra Brown	76%	56.8

- **۲۰ قانون با بالاترین Confidence برای نویسندگان**

- Minimum Support = 0.001
- Minimum Confidence = 0.5

	Condition	Consequent	Confidence	Lift
1	NORA ROBERTS	Nora Roberts	84%	32.82
2	Tim F. Lahaye	Tim Lahaye	78%	127.9
3	MARY HIGGINS CLARK	Mary Higgins Clark	77%	31.39
4	SANDRA BROWN	Sandra Brown	76%	56.8
5	SANDRA BROWN	Nora Roberts	76%	29.77
6	Virginia Henley	Nora Roberts	75%	29.1
7	Candace Camp	Nora Roberts	74%	28.96
8	Tim LaHaye	Tim Lahaye	74%	121
9	Suzanne Brockmann	Nora Roberts	73%	28.65
10	Tim F. Lahaye	Tim LaHaye	73%	269.3
11	Debbie MacOmber	Nora Roberts	72%	28.06
12	Marie Ferrarella	Nora Roberts	71%	27.85
13	Catherine Anderson	Nora Roberts	71%	27.71
14	Jennifer Blake	Nora Roberts	71%	27.64
15	Janelle Taylor	Nora Roberts	71%	27.61
16	Anne Stuart	Nora Roberts	71%	27.59
17	Richard Bachman	Stephen King	71%	16.35
18	Sherryl Woods	Nora Roberts	71%	27.52
19	Kat Martin	Nora Roberts	71%	27.51
20	JoAnn Ross	Nora Roberts	70%	27.27

- **۲۰ قانون با بالاترین LHS Support برای نویسندگان**

	Condition	Consequent	Confidence	Lift	LHS Support
1	Jude Deveraux	Nora Roberts	50%	19.63	0.95%
2	Barbara Delinsky	Nora Roberts	52%	20.42	0.87%
3	DEAN KOONTZ	Stephen King	51%	11.69	0.84%
4	DANIELLE STEEL	Danielle Steel	55%	36.69	0.81%
5	Scott Turow	John Grisham	52%	13.2	0.81%
6	ANNE RICE	Anne Rice	61%	33.95	0.78%
7	Fern Michaels	Nora Roberts	60%	23.3	0.76%
8	John Saul	Stephen King	54%	12.4	0.76%
9	Lawrence Sanders	Stephen King	51%	11.87	0.64%
10	Julie Garwood	Nora Roberts	56%	21.81	0.61%
11	Debbie Macomber	Nora Roberts	59%	22.98	0.61%
12	Richard North Patterson	John Grisham	53%	13.61	0.61%
13	Richard North Patterson	James Patterson	52%	16.8	0.61%
14	Jayne Ann Krentz	Nora Roberts	61%	23.79	0.60%
15	Johanna Lindsey	Nora Roberts	56%	21.77	0.59%
16	Jeffrey Archer	John Grisham	50%	12.74	0.59%
17	Pat Conroy	John Grisham	50%	12.79	0.54%
18	Judith McNaught	Nora Roberts	59%	23.05	0.54%
19	Linda Howard	Nora Roberts	57%	22.35	0.53%
20	Elizabeth Lowell	Nora Roberts	63%	24.62	0.51%